



////AMG

USER MANUAL
MERCEDES AMG
GT3 2020



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DEAR iRACING USER,

Congratulations on your purchase of the Mercedes AMG GT3 2020! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS

**DOUBLE WISHBONE FRONT
AND REAR SUSPENSION**



LENGTH
4710 mm
185.4 in

WIDTH
2040 mm
80.3 in

WHEELBASE
2630 mm
103.5 in

DRY WEIGHT
1320 kg
2910 lbs

WET WEIGHT
WITH DRIVER
1440 kg
3175 lbs

POWER UNIT

6.2L V8 DOHC



DISPLACEMENT
6.2 Liters
378.8 cid

TORQUE
460 lb-ft
625 Nm

POWER
500 bhp
373 kW

RPM LIMIT
7800



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars. To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline.

After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

GETTING STARTED



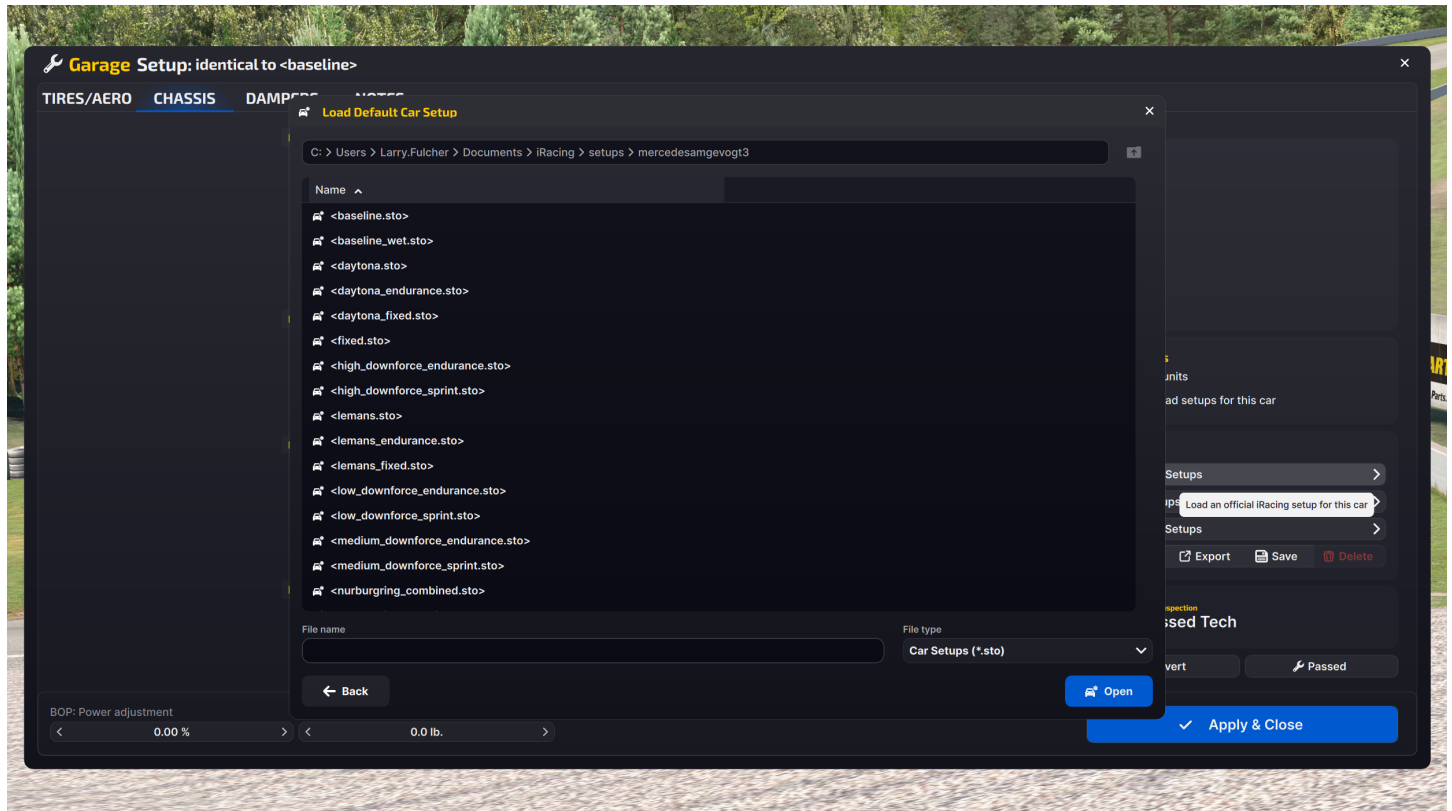
Before starting the car, it is recommended to map controls for Brake Bias, Traction Control and ABS adjustments. While this is not mandatory to drive the car, this will allow you to make quick changes to the driver aid systems to suit your driving style while out on the track.

Once you load into the car, getting started is as easy as selecting the “upshift” button to put it into gear, and hitting the accelerator

pedal. This car uses a sequential transmission and does not require a clutch input to shift in either direction. However the car’s downshift protection will not allow you to downshift if it feels you are traveling too fast for the gear selected and would incur engine damage. If that is the case, the gear change command will simply be ignored.

Upshifting is recommended when all the shift lights flash red, this is at approximately 7025 rpm but will shift up or down slightly depending on the selected gear.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup [baseline.sto]. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply.

If you would like to save your setup for future use click "Save As" on the right to name and save the changes. To access all of your personally saved setups, click "My Setups" on the right side of the garage.

If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH CONFIGURATION

The digital dash display in this car features two selectable pages.

DASH PAGE 1



Top Row	
RPM	Graphical depiction of engine rpm.
Row 2	
T Water	Engine water temperature (Celsius or Fahrenheit)
V Batt	Battery Voltage (V)
Laptime Diff	Lap time delta to best lap
Speed	Road Speed (km/h or mph)
Row 3	
T Oil	Engine oil temperature (Celsius or Fahrenheit)
T Gear	Gearbox oil temperature (Celsius or Fahrenheit)
Lap Time	Last lap time
Gear	Currently selected gear
ABS	Currently selected ABS map
TC	Currently selected Traction Control map
MAP	Currently selected engine map
Fuel	Remaining fuel (Liters or US Gallons)
Bottom Row	
Bottom Bar	Selected dash display page

DASH PAGE 2



Top Row	
RPM	Graphical depiction of engine rpm.
Row 2	
T Water	Engine water temperature (Celsius or Fahrenheit)
V Batt	Battery Voltage (V)
Laptime Diff	Lap time delta to best lap
Speed	Road Speed (km/h or mph)
Row 3	
Top Left	LF air pressure and temperature (Bar or psi / Celsius or Fahrenheit)
Top Right	RF air pressure and temperature (Bar or psi / Celsius or Fahrenheit)
Bottom Left	LR air pressure and temperature (Bar or psi / Celsius or Fahrenheit)
Bottom Right	RR air pressure and temperature (Bar or psi / Celsius or Fahrenheit)
Gear	Currently selected gear
ABS	Currently selected ABS map
TC	Currently selected Traction Control map
MAP	Currently selected engine map
Fuel	Remaining fuel (Liters or US Gallons)
Bottom Row	
Bottom Bar	Selected dash display page

PIT LIMITER



When the pit limiter is active a blue bar will appear at the top of the screen specifying the current vehicle speed. This bar will be blue

while under the limit and red when over. In addition to this, the shift light cluster will flash with alternating blue and yellow lights.

PIT LIMITER



The shift lights illuminate from the outer edges towards the center in the following pattern:

- 2 Green - 6180 rpm
- 4 Green - 6350 rpm
- 2 Yellow - 6520 rpm
- 4 Yellow - 6690 rpm
- 2 Red - 6860 rpm
- All Red Flashing - 7030 rpm

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES & AERO

TIRES

TIRE TYPE:

Tire type: ▼ Dry

LEFT FRONT:

Starting pressure: < 23.0 psi >

Last hot pressure: 23.0 psi

Last temps O M I: 95F 95F 95F

Tread remaining: 100% 100% 100%

RIGHT FRONT:

Starting pressure: < 23.0 psi >

Last hot pressure: 23.0 psi

Last temps I M O: 95F 95F 95F

Tread remaining: 100% 100% 100%

LEFT REAR:

Starting pressure: < 23.0 psi >

Last hot pressure: 23.0 psi

Last temps O M I: 95F 95F 95F

Tread remaining: 100% 100% 100%

RIGHT REAR:

Starting pressure: < 23.0 psi >

Last hot pressure: 23.0 psi

Last temps I M O: 95F 95F 95F

Tread remaining: 100% 100% 100%

AERO BALANCE CALC:

Front RH at speed: < 1.732" >

Rear RH at speed: < 2.638" >

TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

COLD AIR PRESSURE

Air pressure in the tire when the car is loaded into the world. Higher pressures will reduce rolling drag and heat buildup, but will decrease grip. Lower pressures will increase rolling drag and heat buildup, but will increase grip. Higher speeds and loads require higher pressures, while lower speeds and loads will see better performance from lower pressures. Cold pressures should be set to track characteristics for optimum performance. Generally speaking, it is advisable to start at lower pressures and work your way upwards as required.

LAST HOT PRESSURE

Air pressure in the tire after the car has returned to the pits. The difference between cold and hot pressures can be used to identify how the car is progressing through a run in terms of balance, with heavier-loaded tires seeing a larger difference between cold and hot pressures. Ideally, tires that are worked in a similar way should build pressure at the same rate to prevent a change in handling balance over the life of the tire, so cold pressures should be adjusted to ensure that similar tires are at similar pressures once up to operating temperature. Hot pressures should be analyzed once the tires have stabilized after a period of laps. As the number of laps per run will vary depending upon track length a good starting point is approximately 50% of a full fuel run.

TIRE TEMPERATURES

Tire carcass temperatures once the car has returned to the pits. Wheel Loads and the amount of work a tire is doing on-track are reflected in the tire's temperature, and these values can be used to analyze the car's handling balance. Center temperatures are useful for directly comparing the work done by each tire, while the Inner and Outer temperatures are useful for analyzing the wheel alignment (predominantly camber) while on track. These values are measured in three zones across the tread of the tire. Inside, Middle and Outer.

TREAD REMAINING

The amount of tread remaining on the tire once the car has returned to the pits. Tire wear is very helpful in identifying any possible issues with alignment, such as one side of the tire wearing excessively, and can be used in conjunction with tire temperatures to analyze the car's handling balance. These values are measured in the same zones as those of temperature.

AERO BALANCE CALCULATOR

Last hot pressure: 23.0 psi	Last hot pressure: 23.0 psi
Last temps O M I: 95F 95F 95F	Last temps I M O: 95F 95F 95F
Tread remaining: 100% 100% 100%	Tread remaining: 100% 100% 100%

AERO BALANCE CALC:

Front RH at speed: < **1.732"** >
 Rear RH at speed: < **2.638"** >
 Wing setting: < **9 degrees** >
 Front downforce: **44.7%**

To maintain aero balance, for every 1 degree of rear wing angle added:
lower the Front ride height 1.0mm (0.040") OR raise the Rear ride height 2.0mm (0.080").

The Aero Calculator is a tool provided to aid in understanding the shift in aerodynamic balance associated with adjustment of the rear wing setting and front and rear ride heights. It is important to note that the values for front and rear ride height displayed here DO NOT result in any mechanical changes to the car itself, however, changes to the rear wing angle here WILL be applied to the car. This calculator is a reference tool ONLY.

FRONT RH AT SPEED

The Ride Height (RH) at Speed is used to give the Aero Calculator heights to reference for aerodynamic calculations. When using the aero calculator, determine the car's Front Ride height via telemetry at any point on track and input that value into the "Front RH at Speed" setting. It is advisable to use an average value of the LF and RF ride heights as this will provide a more accurate representation of the current aero platform rather than using a single corner height.

REAR RH AT SPEED

The Ride Height (RH) at Speed is used to give the Aero Calculator heights to reference for aerodynamic calculations. When using the aero calculator, determine the car's Rear Ride height via telemetry at any point on track and input that value into the "Front RH at Speed" setting. It is advisable to use an average value of the LR and RR ride heights as this will provide a more accurate representation of the current aero platform rather than using a single corner height.

WING SETTING

The wing setting refers to the relative angle of attack of the rear wing, this is a powerful aerodynamic device which has a significant impact upon the total downforce (and drag!) produced by the car as well as shifting the aerodynamic balance of the car rearwards with increasing angle. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as 'rake'. To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

FRONT DOWNFORCE

This value displays the proportion of downforce acting at the front axle for the given wing and ride height combination set within the calculator parameters. This value is an instantaneous representation of your aero balance at this exact set of parameters and it can be helpful to pick multiple points around a corner or section of track to understand how the aerodynamic balance is moving in differing situations such as braking, steady state cornering and accelerating at corner exit. A higher forwards percentage will result in more oversteer in mid to high speed corners.

CHASSIS

FRONT, BRAKES, LIGHTS

FRONT, BRAKES, LIGHTS:	IN-CAR ADJUSTMENTS:
ARB setting: ▼ D3	Brake pressure bias: < 53.00% >
Total toe-in: < -0.118 in >	ABS setting: < 9 (ABS) >
Front master cyl.: < 0.701 in >	TC setting: < 9 (TC) >
Rear master cyl.: < 0.701 in >	Dash display page: < Page 2 >
Brake pads: ▼ Medium friction	%F WtDist: 48.0%
Endurance lights: ▼ Not Fitted	Cross weight: 50.0%
Night LED strip color: ▼ Blue	
Center front splitter height: 1.992 in	
LEFT FRONT:	RIGHT FRONT:

ARB SETTING

The configuration of the Anti-Roll Bar arms, or “blades”, can be changed to alter the overall stiffness of the ARB assembly. Increasing the setting will increase the roll stiffness of the front suspension, resulting in less body roll but increasing mechanical understeer. This can also, in some cases, lead to a more responsive steering feel from the driver. Conversely, reducing the number of ARB arms will soften the suspension in roll, increasing body roll but decreasing mechanical understeer. This can result in a less-responsive feel from the steering, but grip across the front axle will increase. Along with this, the effects of softening or stiffening the ARB assembly in relation to aerodynamics should also be considered, softer ARB assemblies will result in more body roll which will decrease control of the aero platform in high speed corners and potentially lead to a loss in aero efficiency. The arms can be set to six positions, ranging from D1 (softest) to D6 (stiffest).

TOTAL TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. On the front end, adding toe-out will increase slip in the inside tire while adding toe-in will reduce the slip. Toe-out will decrease straight-line stability but will increase turn-in responsiveness. Toe-in at the front will reduce turn-in responsiveness but will reduce temperature buildup in the front tires.

FRONT MASTER CYLINDER

The Front Brake Master Cylinder size can be changed to alter the line pressure to the front brake calipers. A larger master cylinder will reduce the line pressure to the front brakes, this will shift the brake bias rearwards and increase the pedal effort required to lock the front wheels. A smaller master cylinder will do the opposite and increase brake line pressure to the front brakes, shifting brake bias forward and reducing required pedal effort. 7 Different master cylinder options are available ranging from 15.9 mm / 0.626” (highest line pressure) to 23.8 mm / 0.937” (lowest line pressure).

REAR MASTER CYLINDER

The Rear Brake Master Cylinder size can be changed to alter the line pressure to the rear brake calipers. A larger master cylinder will reduce the line pressure to the rear brakes, this will shift the brake bias forwards and increase the pedal effort required to lock the rear wheels. A smaller master cylinder will do the opposite and increase brake line pressure to the rear brakes, shifting brake bias rearward and reducing required pedal effort. 7 Different master cylinder options are available ranging from 15.9 mm / 0.626” (highest line pressure) to 23.8 mm / 0.937” (lowest line pressure).

BRAKE PADS

The vehicle's braking performance can be altered via the Brake Pad Compound. The “Low” setting provides the least friction, reducing the effectiveness of the brakes, while “Medium” and “High” provide more friction and increase the effectiveness of the brakes while increasing the risk of a brake lockup.

ENDURANCE LIGHT PACKAGE

The vehicle's braking performance can be altered via the Brake Pad Compound. The “Low” setting provides the least friction, reducing the effectiveness of the brakes, while “Medium” and “High” provide more friction and increase the effectiveness of the brakes while increasing the risk of a brake lockup.

NIGHT LED STRIP COLOR

Changes the colour of the LED light strips on the roof of the car.

CENTER FRONT SPLITTER HEIGHT

The front splitter ground clearance is shown in the garage. This is not directly adjustable but will change based on other settings in the garage. There are no tech-inspection checks on this value and it only serves as a reference for the car's attitude under static conditions.



IN-CAR ADJUSTMENTS

FRONT, BRAKES, LIGHTS: ARB setting: ▼ D3 Total toe-in: < -0.118 in > Front master cyl.: < 0.701 in > Rear master cyl.: < 0.701 in > Brake pads: ▼ Medium friction Endurance lights: ▼ Not Fitted Night LED strip color: ▼ Blue Center front splitter height: 1.992 in	IN-CAR ADJUSTMENTS: Brake pressure bias: < 53.00% > ABS setting: < 9 (ABS) > TC setting: < 9 (TC) > Dash display page: < Page 2 > %F WtDist: 48.0% Cross weight: 50.0%
LEFT FRONT: Corner weight: 700.1 lbs	RIGHT FRONT: Corner weight: 700.1 lbs

BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in greater pressure in the front brake line relative to the rear brake line which will shift the brake balance forwards increasing the tendency to lock up the front tyres but potentially increasing overall stability in braking zones. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation. It is important to note that differing combinations of master cylinder size will necessitate differing brake pressure bias values, this is because increasing or reducing the split in master cylinder size difference between front and rear axles will produce an inherent forward or rearward bias in brake line pressure.

ABS SETTING

The current ABS map the car is running. The ABS system features 12 positions divided into three groups to suit varying track conditions, with lower values providing less assistance and higher values providing more assistance to prevent brake lockup. Settings 1-5 are for wet-weather conditions with setting 1 suitable for heavy rain and setting 5 good for lighter rain. Settings 7-11 are for dry conditions on slick tires with reduced support as the setting value increases. Setting 12 disables the system completely.

TRACTION CONTROL SETTING

The position of the traction control switch determines how aggressively the ecu cuts engine torque in reaction to rear wheel spin. 12 positions are available. Settings 1-11 range from least intervention/sensitivity (position 1) through to highest intervention/sensitivity (position 11). Position 12 disables the traction control completely. Position 8 is the recommended baseline setting. More intervention will result in less wheelspin and less rear tire wear but can reduce overall performance if the traction control is cutting engine torque too aggressively and stunting corner exit acceleration.

DASH DISPLAY PAGE

Changes the currently displayed dash page between two available options. Page 1 displays engine and gearbox temperatures while page 2 displays tire pressures and temperatures.

CROSS WEIGHT

The percentage of total vehicle weight in the garage acting across the right front and left rear corners. 50.0% is generally optimal for non-oval tracks as this will produce symmetrical handling in both left and right hand corners providing all other chassis settings are symmetrical. Higher than 50% cross weight will result in more understeer in left hand corners and increased oversteer in right hand corners, cross weight can be adjusted by making changes to the spring perch offsets at each corner of the car.

NOSE WEIGHT

The percentage of total vehicle weight in the garage acting on the front corners. This cannot be adjusted per se but is influenced by the total fuel load carried. As fuel burns (or less starting fuel is specified) the nose weight of the car will increase due to the fuel tank location. This will tend to push the overall balance towards understeer. As such, this reference item can be useful in establishing how much of an adjustment to the setup is required when changing fuel load.

LEFT / RIGHT FRONT

Endurance lights: v Not Fitted	Cross weight: 50.0%
Night LED strip color: v Blue	
Center front splitter height: 1.992 in	
LEFT FRONT:	RIGHT FRONT:
Corner weight: 780 lbs	Corner weight: 780 lbs
Ride height: v 1.994 in v	Ride height: v 1.994 in v
Bump rubber gap: v 0.410 in v	Bump rubber gap: v 0.410 in v
Spring rate: v 1714 lbs/in v	Spring rate: v 1714 lbs/in v
Camber: v -4.0 deg v	Camber: v -4.0 deg v
LEFT REAR:	RIGHT REAR:
Corner weight: 845 lbs	Corner weight: 845 lbs
Ride height: v 3.173 in v	Ride height: v 3.173 in v
Bump rubber gap: v 2.687 in v	Bump rubber gap: v 2.687 in v
Spring rate: v 1571 lbs/in v	Spring rate: v 1571 lbs/in v

CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the spring perch offset adjustments at each corner.

FRONT RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing front ride height will decrease front downforce as well as decrease overall downforce, but will allow for more weight transfer across the front axle when cornering. Conversely, reducing ride height will increase front and overall downforce, but reduce the weight transfer across the front axle.

BUMP RUBBER GAP

The distance the damper will travel before engaging the bump rubber. This will result in a much stiffer suspension and will provide better aerodynamic platform control and better stability in high-speed corners but it will reduce grip in low-speed corners and over rough surfaces. Lower values will engage the bump rubber sooner and higher values will delay engagement to allow for a more compliant suspension.

SPRING RATE

This setting determines the installed corner spring stiffness. Stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control; however, they will also result in increased tire load variation which will manifest as a loss in mechanical grip. Typically the drawbacks of stiffer springs will become more pronounced on rougher tracks and softer springs in these situations will result in increased overall performance. Corner spring changes will influence both roll and pitch control of the platform and ARB changes should be considered when altering corner spring stiffnesses in order to retain the same front to rear roll stiffness and overall balance. When reducing corner spring stiffness the ARB stiffness (either via blade or diameter depending on the size of the corner spring change) should be increased to retain the same roll stiffness as previously. Spring perch offsets must be adjusted to return the car to the prior static ride heights after any spring rate change.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Increasing front camber values will typically result in increased front axle grip during mid to high speed cornering but will result in a loss of braking performance and necessitate a rearward shift in brake bias to compensate.



LEFT / RIGHT REAR

Camber: < -4.0 deg >		Camber: < -4.0 deg >	
LEFT REAR:		RIGHT REAR:	
Corner weight:	845 lbs	Corner weight:	845 lbs
Ride height:	< 3.173 in >	Ride height:	< 3.173 in >
Bump rubber gap:	< 2.687 in >	Bump rubber gap:	< 2.687 in >
Spring rate:	< 1571 lbs/in >	Spring rate:	< 1571 lbs/in >
Camber:	< -3.3 deg >	Camber:	< -3.3 deg >
Toe-in:	< +0.061 in >	Toe-in:	< +0.061 in >
REAR:		DIFFERENTIAL:	

RIDE HEIGHT

Distance from ground to a reference point on the rear of the chassis. Increasing rear ride height will decrease rear downforce as well as increase overall downforce (to a point) and will allow for more weight transfer across the rear axle when cornering. Conversely, reducing ride height will increase rear downforce percentage but reduce overall downforce while reducing the weight transfer across the rear axle. Rear ride height is a critical tuning component for both mechanical and aerodynamic balance considerations and static rear ride heights should be considered and matched to the chosen rear corner springs for optimal performance.

BUMP RUBBER GAP

The distance the damper will travel before engaging the bump rubber. This will result in a much stiffer suspension and will provide better aerodynamic platform control and better stability in high-speed corners but it will reduce grip in low-speed corners and over rough surfaces. Lower values will engage the bump rubber sooner and higher values will delay engagement to allow for a more compliant suspension.

SPRING RATE

Similar to at the front axle, stiffer springs will result in a smaller variance in ride height between high and low load cases and will produce superior aerodynamic performance through improved platform control at the expense of mechanical grip. This can be particularly prominent when exiting slow speed corners with aggressive throttle application. Stiffer springs will tend to react poorly during these instances especially so on rough tracks which will result in significant traction loss. Spring stiffness should be matched to the needs of the racetrack and set such that the handling balance is consistent between high and low speed cornering. As an example case, a car which suffers from high speed understeer but low speed oversteer could benefit from an increase in rear spring stiffness. This will allow for a lower static rear height which will reduce rear weight transfer during slow speed cornering while maintaining or even increasing the rear ride height in high speed cornering to shift the aerodynamic balance forwards and reduce understeer. Spring perch offsets must be adjusted to return the car to the prior static ride heights after any spring rate change.

CAMBER

As at the front of the car it is desirable to run significant amounts of negative camber in order to increase the lateral grip capability; however, it is typical to run slightly reduced rear camber relative to the front. This is primarily for two reasons, firstly, the rear tires are 25 mm (~1") wider compared to the fronts and secondly the rear tires must also perform the duty of driving the car forwards where benefits of camber to lateral grip become a tradeoff against reduced longitudinal (traction) performance.

TOE-IN

At the rear of the car it is typical to run toe-in. Increases in toe-in will result in improved straight line stability and a reduction in response during direction changes. Large values of toe-in should be avoided if possible as this will increase rolling drag and reduce straight line speeds. When making rear toe changes remember that the values are for each individual wheel as opposed to paired as at the front. This means that individual values on the rear wheels are twice as powerful as the combined adjustment at the front of the car when the rear toes are summed together. Generally, it is advised to keep the left and right toe values equal to prevent crabbing or asymmetric handling behavior; however, heavily asymmetric tracks such as Lime Rock Park may see a benefit in performance from running asymmetric configurations of rear toe and other setup parameters.

REAR

REAR:	DIFFERENTIAL:
Fuel level: < 14.0 gal >	Diff Friction Faces: < 8 >
RARB rate: ▾ D4	Diff preload: < 55 ft-lbs >
Wing angle: < 9 degrees >	

FUEL LEVEL

The amount of fuel in the fuel tank when the car is loaded into the world.

RARB RATES

The configuration of the Anti-Roll Bar arms, or “blades”, can be changed to alter the overall stiffness of the ARB assembly. Increasing the ARB assembly stiffness will increase the roll stiffness of the rear suspension, resulting in less body roll but increasing mechanical oversteer. This can also cause the car to “take a set” more quickly at initial turn-in. Conversely, reducing the ARB assembly stiffness will soften the suspension in roll, increasing body roll but decreasing mechanical oversteer. This can result in a less-responsive feel from the rear especially in transient movements, but grip across the rear axle will increase. 7 configurations of ARB arms are available and range from D1 (softest) to D7 (stiffest).

DIFFERENTIAL

REAR:	DIFFERENTIAL:
Fuel level: < 14.0 gal >	Diff Friction Faces: < 8 >
RARB rate: ▾ D4	Diff preload: < 55 ft-lbs >
Wing angle: < 9 degrees >	

DIFF FRICTION PLATES

The number of clutch faces affect how much overall force is applied to keep the differential locked. Treated as a multiplier, adding more faces produces increasingly more locking force but has no impact around zero input torque. This can be considered to be a coarse adjustment to the differential and is most impactful under true coast and wide open throttle situations.

WING ANGLE

The wing setting refers to the relative angle of attack of the rear wing, this is an aerodynamic device which has a significant impact upon the total downforce (and drag!) produced by the car as well as shifting the aerodynamic balance of the car rearwards with increasing angle. Increasing the rear wing angle results in more total cornering grip capability in medium to high speed corners but will also result in a reduction of straight line speed. Rear wing angle should be adjusted in conjunction with front and rear ride heights, specifically the difference between front and rear ride heights known as ‘rake’. To retain the same overall aerodynamic balance it is necessary to increase the rake of the car when increasing the rear wing angle.

DIFF PRELOAD

Diff preload is a static amount of locking force present within the differential and remains constant during both acceleration and deceleration. Increasing diff preload will increase locking on both sides of the differential which will result in more understeer when off throttle and more snap oversteer with aggressive throttle application. Increasing the diff preload will also smooth the transition between on and off throttle behavior as the differential locking force will never reach zero which can be helpful in reducing lift-off oversteer and increasing driver confidence. Typically diff preload should be increased when there is noticeable loss in slow corner exit drive and/or over-rotation during transition between the throttle and brake in low to mid speed corners.

DAMPERS

FRONT DAMPERS

HASSIS	DAMPERS	NOTES
LEFT FRONT:		
LS comp damping: < 5 clicks >		
HS comp damping: < 5 clicks >		
LS rbd damping: < 6 clicks >		
HS rbd damping: < 4 clicks >		
LEFT REAR:		
LS comp damping: < 5 clicks >		
HS comp damping: < 5 clicks >		
LS rbd damping: < 6 clicks >		
RIGHT FRONT:		
LS comp damping: < 5 clicks >		
HS comp damping: < 5 clicks >		
LS rbd damping: < 6 clicks >		
HS rbd damping: < 4 clicks >		
RIGHT REAR:		
LS comp damping: < 5 clicks >		
HS comp damping: < 5 clicks >		
LS rbd damping: < 6 clicks >		

LS COMP DAMPING

Low speed compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. In this case 0 is minimum damping (least resistance to compression) while 11 is maximum damping (most resistance to compression). Increasing the low speed compression damping will result in a faster transfer of weight to this corner of the car during transient movements such as braking and direction change with increased damping usually providing an increase in turn-in response but a reduction in overall grip in the context of front dampers.

HS COMP DAMPING

High speed compression affects the shock's behavior in high speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track. At smoother tracks more high speed compression damping will typically increase performance while at rougher tracks or ones with aggressive kerbs less high speed compression damping can result in an increase in mechanical grip at the expense of platform control. 11 is maximum damping while 0 is minimum damping.

LS RBD DAMPING

Low speed rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't expand enough to maintain proper contact with the track. When tuning for handling, higher front low speed rebound can increase on-throttle mechanical understeer (but reduce nose lift) while lower values will maintain front end grip longer, helping to reduce understeer, but will allow more splitter lift. Excessive front rebound can lead to unwanted oscillations due to the wheel bouncing off of the track surface instead of staying in contact. 11 is maximum damping (most resistant to extension) while 0 is minimum damping (least resistance to extension).

HS RBD DAMPING

High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly. 11 is maximum damping while 0 is minimum damping.

REAR DAMPERS

HS comp damping: < 5 clicks >	HS comp damping: < 5 clicks >
LS rbd damping: < 6 clicks >	LS rbd damping: < 6 clicks >
HS rbd damping: < 4 clicks >	HS rbd damping: < 4 clicks >
LEFT REAR:	RIGHT REAR:
LS comp damping: < 5 clicks >	LS comp damping: < 5 clicks >
HS comp damping: < 5 clicks >	HS comp damping: < 5 clicks >
LS rbd damping: < 6 clicks >	LS rbd damping: < 6 clicks >
HS rbd damping: < 5 clicks >	HS rbd damping: < 5 clicks >

LS COMP DAMPING

Low speed compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. In this case 0 is minimum damping (least resistance to compression) while 11 is maximum damping (most resistance to compression). Increasing the low speed compression damping will result in a faster transfer of weight to this corner of the car during transient movements such as braking and direction change with increased damping usually increasing the cars tendency to understeer on throttle application.

HS COMP DAMPING

High speed compression affects the shock's behavior in high speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track. At smoother tracks more high speed compression damping will typically increase performance while at rougher tracks or ones with aggressive kerbs less high speed compression damping can result in an increase in mechanical grip at the expense of platform control. 11 is maximum damping while 0 is minimum damping.

LS RBD DAMPING

Low speed rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. As at the front, high rebound stiffness will result in improved platform control for aerodynamic performance and overall chassis response but it is important to avoid situations where the shock is too slow in rebounding as this can result in the tire losing complete contact with the track surface. Provided this is avoided,, an increase in rebound stiffness can help to 'slow down' the change in pitch of the car as the brakes are applied, increasing braking stability and off-throttle mechanical understeer. 11 is maximum damping (most resistant to extension) while 0 is minimum damping (least resistance to extension).

HS RBD DAMPING

High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly. 11 is maximum damping while 0 is minimum damping.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.



SETUP TIPS

If the setup fails tech inspection, it is likely the ride heights require adjustment. This is performed by using the spring perch offsets at either end of the car. Right clicks (positive) will reduce the ride height while left clicks (negative) will increase the ride height.

In the iRacing Setups folder you will find a variety of setups.

Baseline is a 100% fuel load setup which is intended solely for loading the car at every possible track. As such, this setup should always pass tech inspection at every fuel load and track but will not provide ultimate performance.

Setups labeled ‘_wet’ have wet tyres pre-fitted and setup adjustments to suit wet conditions.

Setups labeled ‘_sprint’ have a 50% fuel load, a more aggressive balance and are intended for use where there is either a fuel

limitation OR race lengths are approximately 25 to 30 minutes in length. These setups are intended to be used in competition.

Setups labeled ‘_endurance’ have a 100% fuel load and are for use where no fuel restriction is present and/or race lengths are approximately 1 hour or more in length.

The setup titled ‘fixed’ is the setup used in the fixed setup series and is similar to the high_downforce_sprint setup.

Setups labeled ‘nurburgring_’ are built with 70 mm minimum ride heights and are for use solely on Nürburgring Nordschleife configurations.

While most tracks will trend towards favoring more downforce there can be some instances where reducing rear wing angle for less drag may be beneficial. As a rough guide, you can expect the following downforce trims at the following tracks:

Tracks	Downforce Level	Tracks	Downforce Level
Autodromo Jose Carlos Pace	High/Medium	Long Beach Street Circuit	High
Autodromo Nazionale Monza	Medium	Motorsports Arena Oschersleben	High
Brands Hatch Circuit	High	Mount Panorama Circuit	High/Medium
Circuit de Barcelona Catalunya	High	Nürburgring Grand-Prix-Strecke	High
Circuit de Nevers Magny-Cours	High/Medium	Okayama International Circuit	High
Circuit de Spa-Francorchamps	Medium	Road America	High/Medium
Circuit des 24 Heures Du Mans	Medium	Sebring International Raceway	High
Daytona International Speedway	Low/Medium	Silverstone Circuit	High/Medium
Detroit Grand Prix at Belle Isle	High	Sonoma Raceway	High
Fuji International Speedway	High/Medium	Virginia International Raceway	High/Medium
Hungaroring	High	Watkins Glen International	High/Medium
Indianapolis Motor Speedway	Medium	WeatherTech Raceway at Laguna Seca	High
Lime Rock Park	High		

Should you wish to drive at a track not listed it is recommended to start out with the High Downforce setup first before evaluating the other downforce level options. A good indicator of if a track may benefit from a reduction in downforce trim is the maximum speed reached.

The following boundaries are suggestions for what trim level may be optimal but please note that other factors such as track design (number of high speed corners, etc), altitude and ambient conditions will also impact your decision here with higher altitude tracks and hotter ambient conditions favoring more downforce.

Speed	Downforce Level
Max Speed under 250 km/h (155 mph)	High Downforce
Max Speed 250 to 270 km/h	Medium
Max Speed over 270 km/h (167 mph)	Low to Minimum Downforce



AERODYNAMIC TARGETS AND ADJUSTMENTS

GT3 cars are very sensitive to small variations in ride heights at both the front and rear axle and this must be kept in mind when making setup adjustments such as static ride heights, corner spring rates and rear wing angle.

The optimal configuration for most total downforce is as follows:

- Rear Wing Angle: +9
- Dynamic Front Ride Height: 40.0 mm (+/-2.5 mm)
- Dynamic Rear Ride Height: 67.5 mm (+/-2.5 mm)

Should you go over or under the ride height targets stated above you will begin to lose overall downforce. It is very important to consider all aspects of the track when aiming for this maximum downforce target. Consider that if the rear ride height increases beyond the target during braking, you will experience both a balance shift forwards and a loss in overall downforce resulting in a destabilizing situation. It is these braking considerations that will govern how closely you can approach this maximum in a real world situation.

The optimal configuration for the least total drag is as follows:

- Rear Wing Angle: -1
- Dynamic Front Ride Height: 17.5 mm (+/- 2.5 mm)
- Dynamic Rear Ride Height: 17.5 mm (+/- 2.5 mm)

For the majority of tracks, it will be difficult to achieve ride heights low enough to hit these drag targets; however, it is possible at a track such as Daytona. Please keep in mind that your absolute minimums are governed by the road surface and that while aerodynamic drag will decrease as you approach these targets, overall drag may increase if the car starts to make ground contact. It should also be stated that this low drag trim is neither optimal for total downforce nor handling balance.

CHASSIS ADJUSTMENTS

Should you wish to adjust the underpinning balance of the car without impacting the aero platform significantly in pitch and heave, or adjusting the differential then front and rear adjustable anti-roll bars are available.

Stiffer front ARB → More Understeer

Softer front ARB → More Oversteer

Stiffer rear ARB → More Oversteer

Softer rear ARB → More Understeer

When adjusting the rear wing angle, the following adjustments should be made to retain aerodynamic balance:

Rear Wing Angle: +1

Front Ride Height: -1.2 mm

OR

Rear Ride Height: +3.6 mm

Rear Wing Angle: -1

Front Ride Height: +1.2 mm

OR

Rear Ride Height: -3.6 mm

It is also possible to combine adjustments of front and rear ride height together if necessary (such as when lower rear heights cannot be easily achieved), this can result in more overall downforce being retained when reducing wing angle without detrimentally impacting the balance but at the cost of slightly increased aerodynamic drag.

These reference values are provided as targets to aim for, however, overall car balance should remain the priority. It may not be possible to achieve a good balance at these targets in certain situations and as such, you should elect to sacrifice some raw performance for a better balance.

Lower Rear Wing Angle = More oversteer, less downforce, less drag, lower cornering speed, higher straight line speed.

Higher Rear Wing Angle = More understeer, more downforce, more drag, higher cornering speed, lower straight line speed

Softer front AND rear ARB → Reduced aerodynamic performance, more mechanical grip (good for rough surfaces) and slower response to inputs.

Stiffer front AND rear ARB → Increased aerodynamic performance (good for fast sweeping corners), less mechanical grip and increased response to inputs.

DIFFERENTIAL ADJUSTMENTS

Two adjustment options are available for the differential.

More friction faces → More off throttle understeer, more on throttle oversteer, less inside wheelspin-up on rough surfaces and kerb strikes.

Less friction faces → Less off throttle understeer, less on throttle oversteer, more inside wheelspin-up on rough surfaces and kerb strikes. Typically better at tracks like Spa or those with smooth surfaces and flat kerbing.

Friction faces are dominant at high input torques such as full throttle, sustained braking or pure coastdown.

Preload is additive to the total locking torque of the differential and acts as an offset torque which is always present, even at zero input torque. This means that it is more dominant during transition behavior where the differential input torque is near zero, such as at throttle lift and/or during initial trail braking.

More preload → Less liftoff oversteer, more corner entry stability, more off throttle understeer, more on throttle oversteer.

Less preload → More liftoff oversteer, less corner entry stability, less off throttle understeer, less on throttle oversteer.